

Qatar-1b

R-0589

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_241017 · created 2026-07-24T22:31:58+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460600.6759 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.025
Transit depth (Rp/Rs)^2	3.67 +/- 0.96 [%]
Orbital Inclination (inc)	81.97 +/- 0.82
Airmass coefficient 1 (a1)	1.06 +/- 0.022
Airmass coefficient 2 (a2)	-0.042 +/- 0.028
Scatter in the residuals of the lightcurve fit is	2.03 %
Best Comparison Star	#3 - [349, 99]
Optimal Aperture	2.58
Optimal Annulus	12.25
Transit Duration (day)	0.058 +/- 0.0079

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.192 ± 0.025 vs 0.1463 (deviation 1.66σ)
Duration fit vs published	0.058 d vs 0.0692 d (deviation 1.28σ)
Mid-time O–C	-3.4 min
Depth SNR	7.0
Residual scatter	2.03 %

Light curve

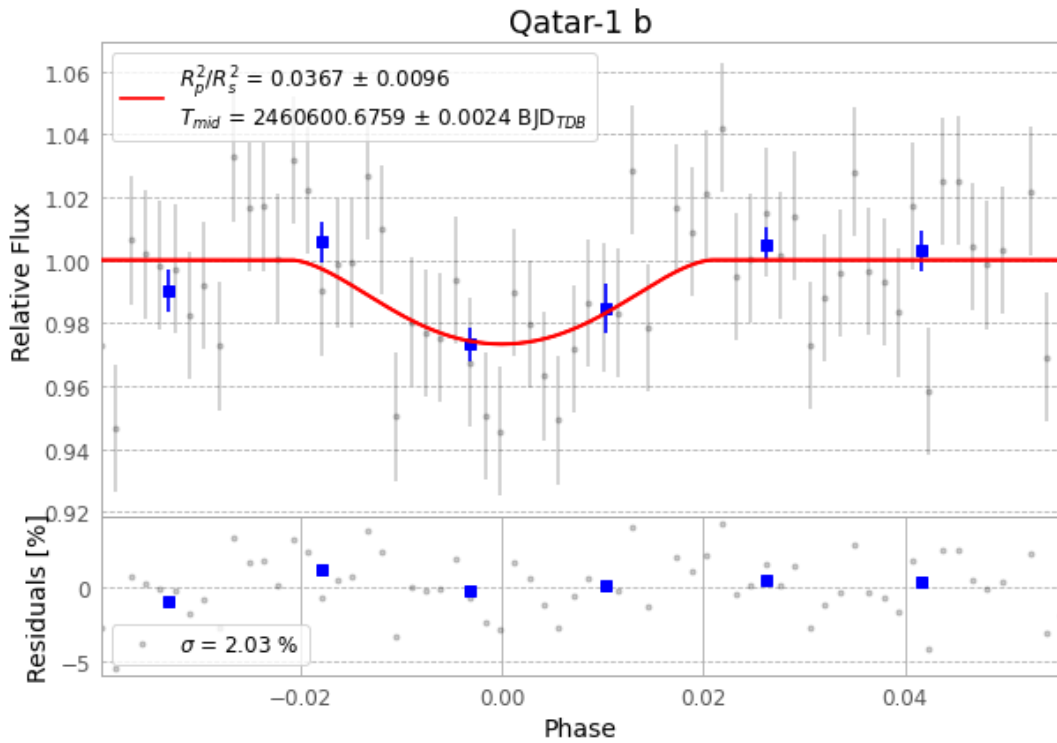


Figure 1 — FinalLightCurve_Qatar-1 b_2024-10-16.png (SHA-256 b86d9a3b7e7283e1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [373, 254], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5f32fa1fba9983d4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

66 FITS frames (43 MB), Qatar-1241017024815.FITS → Qatar-1241017060315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-241017.log (0d771e66bb6c...), FinalLightCurve_Qatar-1 b_2024-10-16.png (b86d9a3b7e72...), FinalLightCurve_Qatar-1 b_2024-10-16.csv (de9f9be29a048...)

Compute resources

Wall time 125.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-24 22:31Z.